CONCEPTUAL MODEL

Reconnecting Dunhuang

The Library Cave and Its Manuscripts

Final Class and Relation Mapping

CIDOC CRM 7.1.3 + CRMsci + CRMdig + minimal project-specific extensions

Status: 15/15 entity classes confirmed · R1–R16 relation mappings locked

1. Purpose and Modelling Strategy

This document consolidates the final Conceptual Model for Reconnecting Dunhuang. It refactors the 15 entities and 16 normalized relations of the Theoretical Model into formal classes and properties while preserving the wording, direction, evidence status, and narrative logic of the original model.

The Conceptual Model follows a reuse-first strategy: CIDOC CRM 7.1.3 is the core ontology; official CRM-family extensions are reused where the core model lacks the necessary precision; project-specific dm: terms are introduced only when no existing term adequately captures the intended semantics.

- Core ontology: CIDOC CRM 7.1.3.
- Extensions: CRMsci for discovery/encounter semantics (R5); CRMdig v5.0 for digitization and digital-object provenance (R14–R16).
- Custom project namespace dm: only for semantics not adequately represented by existing terms: partOfNetwork, locatedNear, providesDigitalAccessTo, virtuallyBringsTogether.
- Event-centric modelling is used for actions such as sealing, discovery, purchase, transfer, and digitization; static relations are not unnecessarily expanded into events.
- Theoretical Model relations remain the source claims. Where Conceptual Model refactoring changes the immediate target (notably R15–R16), the source relation and the refactored implementation are explicitly distinguished.
- Evidence discipline is preserved: R1–R15 are Direct; R16 is Enriched. No unsupported dates, prices, ownership transfers, custody transfers, or institution-specific digitization events are added.

2. Final Entity Class Mapping

All 15 entity instances are mapped to CIDOC CRM classes. No custom class is required for the original entity set.

No.	Entity	Theoretical Type	Conceptual Class	Status
1	Dunhuang Manuscripts	Documentary Collection	crm:E78_Curated_Holding	Confirmed
2	Library Cave (Cave 17)	Place / Heritage Site	crm:E53_Place	Confirmed
3	Mogao Caves	Place / Heritage Site	crm:E53_Place	Confirmed
4	Dunhuang	Place	crm:E53_Place	Confirmed
5	Wang Yuanlu	Person	crm:E21_Person	Confirmed
6	Aurel Stein	Person	crm:E21_Person	Confirmed
7	Paul Pelliot	Person	crm:E21_Person	Confirmed
8	Ōtani Kōzui	Person	crm:E21_Person	Confirmed
9	Silk Road	Historical / Cultural Network	crm:E53_Place	Confirmed
10	Luo Zhenyu	Person	crm:E21_Person	Confirmed
11	British Library	Institution	crm:E40_Legal_Body	Confirmed
12	Bibliothèque nationale de France (BnF)	Institution	crm:E40_Legal_Body	Confirmed
13	National Library of China (NLC)	Institution	crm:E40_Legal_Body	Confirmed
14	Tokyo National Museum	Institution	crm:E40_Legal_Body	Confirmed
15	International Dunhuang Project (IDP)	Digital Heritage Project / Collaborative Initiative	crm:E74_Group	Confirmed

Key class decisions

- Dunhuang Manuscripts = crm:E78_Curated_Holding at aggregate collection level; the model does not decompose the holding into institution-specific sub-holdings.
- Silk Road = crm:E53_Place because the current model treats it as the spatial extent of a historical route network, not the totality of historical exchange activities. It is not retyped as E4 Period merely to obtain a convenient property.
- IDP = crm:E74_Group because it is modelled as a continuing collaborative actor; specific time-bounded activities are represented separately where needed.

3. Final Relation Mapping Overview (R1–R16)

The R numbers below follow Appendix A of the final Theoretical Model and are the sole canonical numbering reference.

R	Canonical source relation	Conceptual formalization	Pattern / target	Ontology	Evidence
R1	Dunhuang → part_of_network → Silk Road	dm:partOfNetwork	E53 Place → E53 Place	dm:	Direct
R2	Mogao Caves → located_near → Dunhuang	dm:locatedNear	E53 Place → E53 Place	dm:	Direct
R3	Library Cave → located_within → Mogao Caves	crm:P89_falls_within	E53 Place → E53 Place	CIDOC CRM	Direct
R4	Dunhuang Manuscripts → sealed_in → Library Cave	Sealing crm:E7_Activity; P12 → Manuscripts; P7 → Library Cave	event-centric	CIDOC CRM	Direct
R5	Wang Yuanlu → discovered → Dunhuang Manuscripts	crmsci:S19_Encounter_Event; P14 → Wang; O19 → Manuscripts; P4 → 25 Jun 1900	event-centric	CRMsci + CRM	Direct
R6	Wang Yuanlu → sold_manuscripts_to → Aurel Stein	crm:E96_Purchase; P23 Wang; P22 Stein; P24 Manuscripts	event-centric	CIDOC CRM	Direct
R7	Wang Yuanlu → sold_manuscripts_to → Paul Pelliot	crm:E96_Purchase; P23 Wang; P22 Pelliot; P24 Manuscripts	event-centric	CIDOC CRM	Direct
R8	Wang Yuanlu → sold_manuscripts_to → Ōtani Kōzui	crm:E96_Purchase; P23 Wang; P22 Ōtani; P24 Manuscripts	event-centric	CIDOC CRM	Direct
R9	Dunhuang Manuscripts → held_by → British Library	crm:P50_has_current_keeper	E78 → E40	CIDOC CRM	Direct
R10	Dunhuang Manuscripts → held_by → BnF	crm:P50_has_current_keeper	E78 → E40	CIDOC CRM	Direct
R11	Dunhuang Manuscripts → held_by → National Library of China	crm:P50_has_current_keeper	E78 → E40	CIDOC CRM	Direct
R12	Dunhuang Manuscripts → held_by → Tokyo National Museum	crm:P50_has_current_keeper	E78 → E40	CIDOC CRM	Direct
R13	Luo Zhenyu → facilitated_transfer_of → Dunhuang Manuscripts	crm:E9_Move; P25 → Manuscripts; P26 → Beijing; P15 → Luo; P4 → 1910	event-centric	CIDOC CRM	Direct
R14	IDP → digitizes → Dunhuang Manuscripts	crmdig:D2_Digitization_Process; P14 → IDP; L1 → Manuscripts	event-centric	CRMdig + CRM	Direct
R15	IDP → provides_digital_access_to → Dunhuang Manuscripts	D2 L11 → D1 Digital Surrogate; IDP dm:providesDigitalAccessTo → same D1	refactored target	CRMdig + dm:	Direct
R16	IDP → virtually_brings_together → Dunhuang Manuscripts	IDP dm:virtuallyBringsTogether → same D1 Digital Surrogate	refactored target	dm: + CRMdig	Enriched

4. Detailed Mapping Decisions by Narrative Layer

R1. Network membership

Source relation: Dunhuang → part_of_network → Silk Road

Evidence: Direct

Dunhuang (crm:E53_Place)
→ dm:partOfNetwork → Silk Road (crm:E53_Place)

No CIDOC CRM 7.1.3 spatial property accurately represents membership in a historical/cultural route network. P89 is spatial containment and would overstate the intended semantics. dm:partOfNetwork is therefore an independent custom property, not a subproperty of P89.

R2. Proximity

Source relation: Mogao Caves → located_near → Dunhuang

Evidence: Direct

Mogao Caves (crm:E53_Place)
→ dm:locatedNear → Dunhuang (crm:E53_Place)

"Near" is a qualitative proximity relation rather than containment or formal topology. A minimal custom property preserves the intended meaning without forcing an inappropriate CRM spatial property.

R3. Spatial containment

Source relation: Library Cave (Cave 17) → located_within → Mogao Caves

Evidence: Direct

Library Cave (crm:E53_Place)
→ crm:P89_falls_within → Mogao Caves (crm:E53_Place)

P89 provides the required spatial-containment semantics and is therefore reused directly.

R4. Sealing in the Library Cave

Source relation: Dunhuang Manuscripts → sealed_in → Library Cave (Cave 17)

Evidence: Direct

Sealing of the Library Cave (crm:E7_Activity)
→ crm:P12_occurred_in_the_presence_of → Dunhuang Manuscripts (crm:E78_Curated_Holding)
→ crm:P7_took_place_at → Library Cave (crm:E53_Place)

The theoretical shortcut sealed_in is refactored as an activity. The model intentionally does not hard-code a debated exact sealing date.

R5. Discovery

Source relation: Wang Yuanlu → discovered → Dunhuang Manuscripts

Evidence: Direct; date: 25 June 1900

Discovery of the Dunhuang Manuscripts (crmsci:S19_Encounter_Event)
→ crm:P14_carried_out_by → Wang Yuanlu (crm:E21_Person)
→ crmsci:O19_encountered_object → Dunhuang Manuscripts (crm:E78_Curated_Holding)
→ crm:P4_has_time-span → 25 June 1900 (crm:E52_Time-Span)

CRMsci is reused because core CRM P12 would only state presence and would not preserve the semantic role of the encountered/discovered object.

R6. Sale / purchase

Source relation: Wang Yuanlu → sold_manuscripts_to → Aurel Stein

Evidence: Direct

Purchase involving Aurel Stein (crm:E96_Purchase)
→ crm:P23_transferred_title_from → Wang Yuanlu (crm:E21_Person)
→ crm:P22_transferred_title_to → Aurel Stein (crm:E21_Person)
→ crm:P24_transferred_title_of → Dunhuang Manuscripts (crm:E78_Curated_Holding)

E96 Purchase is more precise than generic E8 Acquisition for a sale. R6, R7, and R8 are three independent purchase events; combining the buyers into one event would incorrectly imply a joint purchase. No price, currency, or date is added because Appendix A does not provide them for these relations.

R7. Sale / purchase

Source relation: Wang Yuanlu → sold_manuscripts_to → Paul Pelliot

Evidence: Direct

Purchase involving Paul Pelliot (crm:E96_Purchase)
→ crm:P23_transferred_title_from → Wang Yuanlu (crm:E21_Person)
→ crm:P22_transferred_title_to → Paul Pelliot (crm:E21_Person)
→ crm:P24_transferred_title_of → Dunhuang Manuscripts (crm:E78_Curated_Holding)

E96 Purchase is more precise than generic E8 Acquisition for a sale. R6, R7, and R8 are three independent purchase events; combining the buyers into one event would incorrectly imply a joint purchase. No price, currency, or date is added because Appendix A does not provide them for these relations.

R8. Sale / purchase

Source relation: Wang Yuanlu → sold_manuscripts_to → Ōtani Kōzui

Evidence: Direct

Purchase involving Ōtani Kōzui (crm:E96_Purchase)
→ crm:P23_transferred_title_from → Wang Yuanlu (crm:E21_Person)
→ crm:P22_transferred_title_to → Ōtani Kōzui (crm:E21_Person)
→ crm:P24_transferred_title_of → Dunhuang Manuscripts (crm:E78_Curated_Holding)

E96 Purchase is more precise than generic E8 Acquisition for a sale. R6, R7, and R8 are three independent purchase events; combining the buyers into one event would incorrectly imply a joint purchase. No price, currency, or date is added because Appendix A does not provide them for these relations.

R9. Current institutional holding

Source relation: Dunhuang Manuscripts → held_by → British Library

Evidence: Direct

Dunhuang Manuscripts (crm:E78_Curated_Holding)
→ crm:P50_has_current_keeper → British Library (crm:E40_Legal_Body)

P50 is preferred because held_by supports current custody/keeping, not necessarily legal ownership (P52) or the stronger curatorial-responsibility semantics of P109. No E10 Transfer of Custody event is invented because the source relation states the current holding condition rather than a transfer process.

R10. Current institutional holding

Source relation: Dunhuang Manuscripts → held_by → Bibliothèque nationale de France (BnF)

Evidence: Direct

Dunhuang Manuscripts (crm:E78_Curated_Holding)
→ crm:P50_has_current_keeper → Bibliothèque nationale de France (BnF) (crm:E40_Legal_Body)

P50 is preferred because held_by supports current custody/keeping, not necessarily legal ownership (P52) or the stronger curatorial-responsibility semantics of P109. No E10 Transfer of Custody event is invented because the source relation states the current holding condition rather than a transfer process.

R11. Current institutional holding

Source relation: Dunhuang Manuscripts → held_by → National Library of China

Evidence: Direct

Dunhuang Manuscripts (crm:E78_Curated_Holding)
→ crm:P50_has_current_keeper → National Library of China (crm:E40_Legal_Body)

P50 is preferred because held_by supports current custody/keeping, not necessarily legal ownership (P52) or the stronger curatorial-responsibility semantics of P109. No E10 Transfer of Custody event is invented because the source relation states the current holding condition rather than a transfer process.

R12. Current institutional holding

Source relation: Dunhuang Manuscripts → held_by → Tokyo National Museum

Evidence: Direct

Dunhuang Manuscripts (crm:E78_Curated_Holding)
→ crm:P50_has_current_keeper → Tokyo National Museum (crm:E40_Legal_Body)

P50 is preferred because held_by supports current custody/keeping, not necessarily legal ownership (P52) or the stronger curatorial-responsibility semantics of P109. No E10 Transfer of Custody event is invented because the source relation states the current holding condition rather than a transfer process.

R13. Facilitated transfer to Beijing

Source relation: Luo Zhenyu → facilitated_transfer_of → Dunhuang Manuscripts

Evidence: Direct; scope: most remaining Chinese manuscripts; transfer to Beijing, 1910

Transfer of the Dunhuang Manuscripts to Beijing (crm:E9_Move)
→ crm:P25_moved → Dunhuang Manuscripts (crm:E78_Curated_Holding)
→ crm:P26_moved_to → Beijing (crm:E53_Place)
→ crm:P15_was_influenced_by → Luo Zhenyu (crm:E21_Person)
→ crm:P4_has_time-span → 1910 (crm:E52_Time-Span)

E9 Move captures the evidenced physical relocation to Beijing without asserting a custody or ownership transfer. P15 preserves the cautious semantics of facilitated: it records Luo Zhenyu as having a bearing on the move without asserting that he directly carried it out or bore legal responsibility. P27 moved from is omitted because the Appendix A note does not state an origin.

R14. Digitization

Source relation: International Dunhuang Project → digitizes → Dunhuang Manuscripts

Evidence: Direct

Digitization of the Dunhuang Manuscripts (crmdig:D2_Digitization_Process)
→ crm:P14_carried_out_by → International Dunhuang Project (crm:E74_Group)
→ crmdig:L1_digitized → Dunhuang Manuscripts (crm:E78_Curated_Holding)

CRMdig is reused because D2 and L1 directly model the digitization process and the physical thing digitized.At the R14 stage alone, no digital-output node is required; the digital surrogate is introduced in R15 and linked retrospectively to the R14 digitization process through crmdig:L11_had_output.

R15. Digital access

Source relation: International Dunhuang Project → provides_digital_access_to → Dunhuang Manuscripts

Evidence: Direct

Digitization of the Dunhuang Manuscripts (crmdig:D2_Digitization_Process)
→ crmdig:L11_had_output → Digital Surrogate of the Dunhuang Manuscripts (crmdig:D1_Digital_Object)

International Dunhuang Project (crm:E74_Group)
→ dm:providesDigitalAccessTo → the same crmdig:D1_Digital_Object

The source relation is preserved verbatim at the Theoretical Model layer. In the Conceptual Model, "digital access" is operationalized through the digital surrogate produced by digitization. This refactoring makes explicit that online access is to a digital representation, not physical access to the manuscripts themselves. Core CRM and CRMdig do not provide a property with the required access-provision semantics, so a minimal dm: property is introduced.

R16. Virtual reconnection

Source relation: International Dunhuang Project → virtually_brings_together → Dunhuang Manuscripts

Evidence: Enriched; supported by IDP institutional description

International Dunhuang Project (crm:E74_Group)
→ dm:virtuallyBringsTogether
→ Digital Surrogate of the Dunhuang Manuscripts (crmdig:D1_Digital_Object; same node as R15)

R16 is modelled as a conceptual synthesis of R9–R12 (physical dispersal across institutional keepers) and R14–R15 (digitization and digital-access provision). It expresses the virtual reconnection of materials that remain physically dispersed and does not imply physical reunification, ownership transfer, or custody transfer. The `crmdig:D1_Digital_Object` *represents the digital surrogate of the Dunhuang Manuscripts at the aggregate level. This does not imply that all manuscript materials constitute a single indivisible digital file, nor does it assert separate digitization processes for individual holding institutions. Rather, this modelling preserves the abstraction level of the Theoretical Model, which treats the Dunhuang Manuscripts as an aggregate collection. The same D1 node is reused in R15 and R16: R15 represents the provision of digital access, whereas R16 expresses the higher-level virtual reconnection of the dispersed materials. A more granular model could introduce institution-specific digital objects if supported by additional evidence.*

5. Project-Specific Property Definitions

Property	Domain	Range	Symmetric	Transitive	Scope note
dm:partOfNetwork	crm:E53_Place	crm:E53_Place	No	No	Relates a place to a broader historical or cultural route network in which the place functions as a constituent location or node. This property expresses network membership or association rather than geometric spatial containment.
dm:locatedNear	crm:E53_Place	crm:E53_Place	Yes	No	Relates two places that are qualitatively near one another in the spatial interpretation used by the project. It does not imply containment, overlap, or a measurable fixed distance.
dm:providesDigitalAccessTo	crm:E39_Actor	crmdig:D1_Digital_Object	No	No	Relates an actor to a digital object that the actor makes available for digital or online access. It represents provision of access to a digital representation and does not imply access to the physical heritage object itself.
dm:virtuallyBringsTogether	crm:E39_Actor	crmdig:D1_Digital_Object	No	No	Relates an actor to a digital object through which geographically or institutionally dispersed heritage materials are virtually reconnected or presented within a unified digital environment. It expresses virtual rather than physical reunification and does not imply any transfer of ownership or custody.

Important: dm:partOfNetwork is not declared as a subproperty of crm:P89_falls_within. dm:virtuallyBringsTogether and dm:providesDigitalAccessTo are also kept independent; their domain/range shapes overlap, but they assert different semantics.

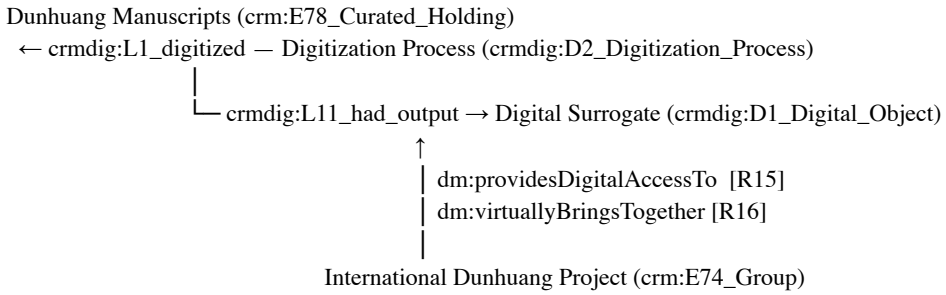
6. Additional Conceptual Nodes Introduced During Formalization

The original Theoretical Model contains 15 entity instances. The Conceptual Model introduces additional event, place, time-span, and digital-object nodes only where formal representation requires them. These are modelling nodes, not replacements for the original entity list.

Node	Class	Used in	Modelling function
Sealing of the Library Cave	crm:E7_Activity	R4	Represents the sealing action.
Discovery of the Dunhuang Manuscripts	crmsci:S19_Encounter_Event	R5	Represents the encounter/ discovery event.
Purchase involving Aurel Stein	crm:E96_Purchase	R6	Independent purchase event.
Purchase involving Paul Pelliot	crm:E96_Purchase	R7	Independent purchase event.
Purchase involving Ōtani Kōzui	crm:E96_Purchase	R8	Independent purchase event.
Transfer to Beijing	crm:E9_Move	R13	Represents physical relocation, not custody/ownership transfer.
Beijing	crm:E53_Place	R13	Destination required by P26 moved to.
1910	crm:E52_Time-Span	R13	Time-span supported by Appendix A scope note.
25 June 1900	crm:E52_Time-Span	R5	Discovery date supported directly.
Digitization of the Dunhuang Manuscripts	crmdig:D2_Digitization_Process	R14–R15	Digitization process.
Digital Surrogate of the Dunhuang Manuscripts	crmdig:D1_Digital_Object	R15–R16	Single aggregate digital-representation node reused by both relations at current project granularity.

7. Digital Reconnection Chain (R14–R16)

The final three relations form a single controlled chain without collapsing their distinct claims:



Interpretive discipline: R14 states digitization; R15 states provision of digital access; R16 states virtual reconnection. R16 remains an Enriched relation and should not be treated as if it came from the same starting source as the Direct relations.

8. Explicitly Rejected / Avoided Mappings

Relation	Rejected option	Reason
R1	crm:P89_falls_within	Rejected because network membership is not geometric containment.
R5	crm:P12_occurred_in_the_presence_of as the sole discovery link	Rejected as semantically too weak; it does not say that the object was encountered/ discovered.
R6–R8	crm:E8_Acquisition only	Rejected as unnecessarily broad when crm:E96_Purchase captures a sale.
R9–R12	crm:P52_has_current_owner	Rejected because held_by does not establish legal ownership.
R9–R12	crm:P109_has_current_or_former_curator	Rejected because curatorial responsibility is stronger than the evidence.
R13	crm:P14_carried_out_by → Luo Zhenyu	Rejected because facilitated does not warrant asserting direct execution or causal/ legal responsibility.
R13	crm:E10_Transfer_of_Custody	Rejected because the evidence states transfer to Beijing, not a specific custody handover from one actor to another.
R15	crm:P76_has_contact_point	Rejected because a URL/contact point is not equivalent to providing online access to heritage material.
R15	crm:P14 / P16 / P67	Rejected because each distorts access-provision semantics.
R16	Physical reunification / Move / custody / ownership transfer	Rejected because R16 describes virtual reconnection while the manuscripts remain physically dispersed.

9. Ontology Stack and Final Status

- CIDOC CRM 7.1.3: primary conceptual backbone for persons, places, organizations, curated holdings, activities, moves, purchases, custody/keeper relations, and temporal qualification.
- CRMsci: reused for R5 through S19 Encounter Event and O19 encountered object, providing a more precise discovery pattern than core CRM alone.
- CRMdig v5.0: reused for R14–R16 through D2 Digitization Process, L1 digitized, L11 had output, and D1 Digital Object.
- dm: project namespace: four minimal properties where no existing ontology term accurately preserves the theoretical semantics.

FINAL STATUS

15/15 entity classes CONFIRMED · R1–R16 relation mappings LOCKED

The next production step is diagram construction from this frozen mapping table, not further semantic redesign.